25900264 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBOTA05 - BIO-INSTRUMENTATION AND QUALITY ANALYSIS 1

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer in One or Two Sentences

Answer any **10** questions

Each question carries **2** marks

1. What is use of hot plate? [K] / [CO1]
2. You are provided 1 M Tris buffer. How will you dilute this to make 500 ml 0.1 M buffer [U] / [CO2]
3. What is the key difference between the equivalence point and the end point in a titration? [K] / [CO3]
4. What are the detectors used in Atomic absorption spectrophotometer? [K] / [CO4]
5. List out the uses of a conical flask. [K] / [CO1]
6. What is the composition of carbonate-bicarbonate buffer? [K] / [CO2]
7. Give any two applications of gravimetry in food analysis. [U] / [CO3]
8. What is the use of a colorimeter? [K] / [CO4]
9. Describe air displacement micropipette? [K] / [CO1]
10. List out any two importance of pKa value [K] / [CO2]

11. Describe the heat transfer mechanism in a laboratory oven [K] / [CO1]

12. What is meant by strong acid? [K] / [CO2]

[2x10 = 20]

Part B

Short Answer Type Questions

Answer any **8** questions

Each question carries **5** marks

13. Recall the concepts of Molarity and ppm. [K] / [CO1]

14. Draw a labelled diagram of pH Meter. [U] / [CO2]

15. Write a brief note on safety protocols to be followed during acid- alkali titration. [K] / [CO3]

16. Explain briefly electromagnetic spectrum. [U] / [CO4]

17. Describe the concept of Distilled water and Ultrapure water. [K] / [CO1]

18. What is a buffer. What are the factors that affect buffering capacity? [K] / [CO2]

19. Give a brief note on methyl orange and bromothymol blue as visual indicators in acid-alkali titrations. [U] / [CO3]

20. What is the principle of a UV-Visible spectrophotometer? [K] / [CO4]

21. What is an electronic balance? What are the precautions to be taken while using an electronic balance? [K] / [CO1]

22. Explain the measurement of pH using litmus paper and pH paper. [U] / [CO2]

[5x8 = 40]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

23. Explain the various steps in gravimetric analysis. [U] / [CO3]

24. Explain the principle of scattering spectroscopy with special reference to working and use of Raman spectrophotometer. [U] / [CO4]

[10x1 = 10]